

Principles of Microeconomics — Day 4

In-Class Problem Set: Duopoly

This problem set accompanies the Day 4 session. Numbers differ from the lecture example. Solutions (instructor copy) are collected at the end; set `\solnfalse` in the preamble to produce a student version.

Exercise 1 — Three ways two firms can compete

Two firms face market demand $P = 100 - Q$ with $Q = q_1 + q_2$, and each has constant marginal cost $MC = 10$.

- (a) **Price competition** (identical product): find the price and total quantity.
- (b) **Quantity competition** (both choose output simultaneously): derive both response functions and solve for equilibrium quantities, price and each firm's profit.
- (c) **First mover** (firm 1 commits first): solve for q_1 , q_2 , price and each firm's profit.
- (d) Tabulate total output, price and industry profit for monopoly, quantity competition, the first-mover case and price competition. Rank total output.

Exercise 2 — Quantity competition with unequal costs

Two firms face market demand $P = 120 - Q$ with $Q = q_1 + q_2$. They choose quantities simultaneously, but their costs differ: firm 1 has $MC_1 = 20$ and firm 2 has $MC_2 = 40$ (no fixed costs).

- (a) Derive each firm's response function.
- (b) Solve for the equilibrium quantities, the market price, and each firm's profit.
- (c) Which firm produces more, and why?
- (d) Now suppose instead the two firms compete on *price*. With identical products, who ends up serving the market, and at roughly what price?